

Robotic Deployment and Site Preparation Level 4 Work Packages

Candidate scope and control definitions version 0.1

Owner	Program Director	Status	Draft for review
Version	0.1	Basis	WBS Dictionary v0.1
Coverage	30 Level 4 work packages	Date	22 September 2026

Purpose

This document decomposes WBS element 1.3 Robotic Deployment and Site Preparation into 30 candidate Level 4 work packages. It converts the Level 3 outcomes into manageable units that can later receive named ownership, schedule dates, resources, budgets, risks, and performance measures.

Program decision requested

Review the proposed scope boundaries, acceptance evidence, accountable functions, dependencies, and gates. Approve selected packages for detailed planning only after branch owners confirm that the decomposition is complete, nonduplicative, and aligned with the program architecture.

Director control intent

Use these work packages to establish accountability and integration without prematurely authorizing execution. Robotic Deployment owns the means and evidence for surveying, preparing, installing, inspecting, and remotely sustaining the site. The Habitat, Lunar Data Center, and Shared Infrastructure branches retain design authority and operational acceptance for their systems.

Work Package Control Framework

Minimum controls required before baseline authorization

Definition of a controlled work package

A work package is the lowest WBS element authorized for integrated cost, schedule, scope, and performance control. A package should produce a measurable outcome, have one accountable owner, contain bounded work, and close with objective evidence accepted at a defined decision gate.

Baseline requirements

- Name one accountable manager and identify the supporting functions.
- Convert the candidate scope into scheduled activities with logical predecessors and successors.
- Assign planned start and finish dates, resources, budget, and measurable completion criteria.
- Link technical requirements, interface-control documents, verification evidence, risks, and configuration items.
- Establish status methods, variance thresholds, forecast expectations, and change-control authority.
- Confirm that safety-critical and crew-readiness decisions remain with the authorized decision bodies.

Control sequence

Step	Program action	Primary evidence	Decision
1	Validate scope	Inclusions exclusions interfaces	Accept revise or remove
2	Plan execution	Activities resources estimates risks	Authorize baseline development
3	Baseline package	Owner budget schedule measures	Authorize controlled work
4	Monitor performance	Status forecast variance evidence	Continue recover or change
5	Close package	Accepted completion evidence	Close and hand over

Level 4 Decomposition Index

Thirty candidate work packages organized by Level 3 parent

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1.3.1.1 Robotic Mission Concept of Operations and Requirements

Parent element 1.3.1 Robotic Fleet and Payload Configuration

Level 4 Work Package Definition	
Work package outcome	Approved robotic mission concept, operating scenarios, and allocated requirements baseline.
Included work	Mission phases, operating modes, task allocation, environmental constraints, safety priorities, command authority, performance requirements, and verification approach.
Excluded work	Detailed vehicle design, supplier selection, launch integration, and site-specific execution procedures.
Acceptance evidence	The concept and allocated requirements are approved, traceable to program objectives, and linked to verification methods.
Key dependencies	Program Charter v1.1, WBS v0.2, system architecture, site assumptions, and requirements from elements 1.4 through 1.6.
Accountable function	Robotics Systems Engineering Lead
Supporting functions	Program Controls, Safety, Habitat, Data Center, Shared Infrastructure
Primary risk	Unclear or conflicting mission requirements drive redesign and schedule instability.
Decision gate	Robotic Fleet Readiness Review
Authorization status	Candidate work package for planning. Budget, schedule, named owner, and execution authority are not yet baselined.

Control note: The accountable function becomes responsible for scope, cost, schedule, risk, interfaces, and completion evidence only after formal baseline authorization.

1.3.1.2 Fleet Architecture and Role Allocation

Parent element 1.3.1 Robotic Fleet and Payload Configuration

Level 4 Work Package Definition	
Work package outcome	Approved fleet composition and role allocation for survey, transport, construction, inspection, maintenance, and contingency tasks.
Included work	Vehicle classes, quantities, capacity margins, task assignments, redundancy strategy, interoperability, operating sequence, and degraded-mode coverage.
Excluded work	Detailed component design, manufacturing planning, and procurement authorization.
Acceptance evidence	Fleet sizing and role allocation close all approved mission tasks with documented capacity, redundancy, and availability margins.
Key dependencies	1.3.1.1, cargo manifest assumptions, site traversability data, communication coverage, and maintenance concept.
Accountable function	Robotic Fleet Architecture Lead
Supporting functions	Systems Engineering, Logistics, Communications, Mission Operations
Primary risk	Insufficient capacity or common-mode failures prevent completion of critical surface tasks.
Decision gate	Robotic Fleet Readiness Review
Authorization status	Candidate work package for planning. Budget, schedule, named owner, and execution authority are not yet baselined.

Control note: The accountable function becomes responsible for scope, cost, schedule, risk, interfaces, and completion evidence only after formal baseline authorization.

1.3.1.3 Payload Tooling and Interface Configuration

Parent element 1.3.1 Robotic Fleet and Payload Configuration

Level 4 Work Package Definition	
Work package outcome	Controlled payload, end-effector, tooling, and interface configuration for each robotic mission role.
Included work	Payload manifests, mechanical and electrical interfaces, tool-change approach, sensing packages, lifting fixtures, calibration needs, and interface-control documentation.
Excluded work	Design ownership of habitat, data-center, or common-infrastructure equipment being handled.
Acceptance evidence	All payload and tooling configurations satisfy mass, power, data, structural, handling, and environmental interface requirements with approved interface evidence.
Key dependencies	1.3.1.2, equipment provider interface data, transportation constraints, and surface assembly plans.
Accountable function	Robotic Payload Integration Lead
Supporting functions	Mechanical, Electrical, Avionics, Habitat, Data Center, Logistics
Primary risk	Late interface changes or incompatible tooling prevent safe handling and assembly.
Decision gate	Robotic Fleet Readiness Review
Authorization status	Candidate work package for planning. Budget, schedule, named owner, and execution authority are not yet baselined.

Control note: The accountable function becomes responsible for scope, cost, schedule, risk, interfaces, and completion evidence only after formal baseline authorization.

1.3.1.4 Mobility Autonomy and Fault Management Verification

Parent element 1.3.1 Robotic Fleet and Payload Configuration

Level 4 Work Package Definition	
Work package outcome	Verified mobility, autonomy, perception, navigation, safe-state, and fault-management capability for representative lunar conditions.
Included work	Simulation, hardware-in-the-loop testing, terrain testing, autonomy logic, obstacle response, loss-of-communication response, fault isolation, recovery, and evidence capture.
Excluded work	Final mission certification and launch-site acceptance testing.
Acceptance evidence	Defined mobility and autonomy scenarios meet performance thresholds, and all safety-critical faults transition to approved safe or recoverable states.
Key dependencies	1.3.1.1 through 1.3.1.3, terrain models, communications architecture, flight software, and hazard controls.
Accountable function	Robotics Verification Lead
Supporting functions	Autonomy, GN&C, Software, Safety, Test, Communications
Primary risk	Unverified edge cases cause loss of mobility, asset damage, or mission interruption.
Decision gate	Robotic Fleet Readiness Review
Authorization status	Candidate work package for planning. Budget, schedule, named owner, and execution authority are not yet baselined.

Control note: The accountable function becomes responsible for scope, cost, schedule, risk, interfaces, and completion evidence only after formal baseline authorization.

1.3.1.5 Integrated Fleet Qualification and Launch Readiness

Parent element 1.3.1 Robotic Fleet and Payload Configuration

Level 4 Work Package Definition	
Work package outcome	Integrated qualification and launch-readiness evidence for the complete robotic fleet and payload set.
Included work	Environmental qualification status, end-to-end mission rehearsals, configuration audits, open-work closure, logistics verification, launch-site support, and readiness recommendation.
Excluded work	Launch vehicle mission assurance and transportation operations controlled by element 1.2.
Acceptance evidence	Qualification evidence is complete, configurations match the controlled manifest, critical discrepancies are closed, and the fleet is accepted for payload integration.
Key dependencies	1.3.1.1 through 1.3.1.4, element 1.2 payload-integration criteria, logistics plan, and program risk disposition.
Accountable function	Robotics Mission Assurance Lead
Supporting functions	Quality, Test, Logistics, Transportation, Program Controls
Primary risk	Late qualification failures or configuration mismatches delay payload integration and launch.
Decision gate	Robotic Fleet Readiness Review
Authorization status	Candidate work package for planning. Budget, schedule, named owner, and execution authority are not yet baselined.

Control note: The accountable function becomes responsible for scope, cost, schedule, risk, interfaces, and completion evidence only after formal baseline authorization.

1.3.2.1 Survey Data Requirements and Campaign Plan

Parent element 1.3.2 Lunar Site Survey and Hazard Characterization

Level 4 Work Package Definition	
Work package outcome	Approved survey campaign defining required data, coverage, resolution, collection sequence, and decision use.
Included work	Stakeholder data needs, survey routes, sensor assignments, calibration, sampling strategy, data quality, prioritization, contingency collection, and campaign schedule.
Excluded work	Final site-selection decision and detailed infrastructure design.
Acceptance evidence	Every site decision criterion is mapped to a planned measurement, required resolution, responsible platform, and acceptance threshold.
Key dependencies	1.3.1 fleet capabilities, candidate-site data, element 1.2 landing constraints, and requirements from elements 1.4 through 1.6.
Accountable function	Site Characterization Lead
Supporting functions	Science, Geotechnical, Mission Planning, Habitat, Data Center, Transportation
Primary risk	Missing or low-quality survey data leaves critical site decisions unsupported.
Decision gate	Lunar Site Characterization Review
Authorization status	Candidate work package for planning. Budget, schedule, named owner, and execution authority are not yet baselined.

Control note: The accountable function becomes responsible for scope, cost, schedule, risk, interfaces, and completion evidence only after formal baseline authorization.

1.3.2.2 Terrain Topography and Geotechnical Characterization

Parent element 1.3.2 Lunar Site Survey and Hazard Characterization

Level 4 Work Package Definition	
Work package outcome	Validated terrain, slope, roughness, regolith, bearing-capacity, and subsurface characterization data.
Included work	Topographic mapping, slope and roughness assessment, regolith sampling, bearing tests, settlement indicators, subsurface anomalies, uncertainty analysis, and data products.
Excluded work	Structural foundation design and execution of site grading.
Acceptance evidence	Terrain and geotechnical datasets meet approved coverage and quality thresholds and identify usable and restricted areas with stated uncertainty.
Key dependencies	1.3.2.1, calibrated survey payloads, positioning services, and candidate-site access.
Accountable function	Lunar Geotechnical Lead
Supporting functions	Robotics, Survey, Structures, Data Management, Safety
Primary risk	Unrecognized weak soil, slopes, or subsurface hazards compromise landing, mobility, or foundations.
Decision gate	Lunar Site Characterization Review
Authorization status	Candidate work package for planning. Budget, schedule, named owner, and execution authority are not yet baselined.

Control note: The accountable function becomes responsible for scope, cost, schedule, risk, interfaces, and completion evidence only after formal baseline authorization.

1.3.2.3 Illumination Thermal Radiation and Dust Characterization

Parent element 1.3.2 Lunar Site Survey and Hazard Characterization

Level 4 Work Package Definition	
Work package outcome	Validated environmental profiles supporting power, thermal, radiation, dust, and operational planning.
Included work	Illumination cycles, shadowing, thermal extremes, radiation background, dust properties and transport indicators, sensor observations, uncertainty, and seasonal assumptions.
Excluded work	System-specific shielding, radiator, solar-array, or dust-control design.
Acceptance evidence	Environmental profiles cover the approved operating horizon and provide bounded inputs accepted by the owning system branches.
Key dependencies	1.3.2.1, environmental sensors, orbital datasets, candidate-site geometry, and system input requirements.
Accountable function	Lunar Environment Lead
Supporting functions	Power, Thermal, Radiation, Dust, Habitat, Data Center
Primary risk	Environmental conditions exceed design assumptions or remain too uncertain for safe operations.
Decision gate	Lunar Site Characterization Review
Authorization status	Candidate work package for planning. Budget, schedule, named owner, and execution authority are not yet baselined.

Control note: The accountable function becomes responsible for scope, cost, schedule, risk, interfaces, and completion evidence only after formal baseline authorization.

1.3.2.4 Landing Traverse and Work Zone Hazard Mapping

Parent element 1.3.2 Lunar Site Survey and Hazard Characterization

Level 4 Work Package Definition	
Work package outcome	Controlled hazard maps for landing zones, traverse corridors, construction areas, exclusion zones, and emergency routes.
Included work	Boulder and crater mapping, slope limits, trafficability, plume interaction areas, line-of-sight constraints, work-zone conflicts, safe havens, and map configuration control.
Excluded work	Final landing trajectory design and execution of obstacle removal.
Acceptance evidence	Hazard maps are georeferenced, configuration controlled, validated against collected data, and accepted by transportation and surface-operations stakeholders.
Key dependencies	1.3.2.2, 1.3.2.3, landing-system constraints, robotic mobility limits, and surface-network coverage.
Accountable function	Surface Hazard Analysis Lead
Supporting functions	Transportation, Robotics, Safety, Mission Operations, Communications
Primary risk	Incomplete or outdated hazard maps cause unsafe landing, traversal, or simultaneous operations.
Decision gate	Lunar Site Characterization Review
Authorization status	Candidate work package for planning. Budget, schedule, named owner, and execution authority are not yet baselined.

Control note: The accountable function becomes responsible for scope, cost, schedule, risk, interfaces, and completion evidence only after formal baseline authorization.

1.3.2.5 Site Suitability Recommendation and Baseline

Parent element 1.3.2 Lunar Site Survey and Hazard Characterization

Level 4 Work Package Definition	
Work package outcome	Integrated site-suitability recommendation and approved evidence baseline for program decision making.
Included work	Multi-criteria assessment, alternatives comparison, uncertainty, infrastructure compatibility, risk and opportunity evaluation, stakeholder concurrence, decision record, and baseline release.
Excluded work	Political, treaty, ownership, or final external authorization decisions beyond program control.
Acceptance evidence	The recommendation traces to approved criteria, documents residual risks and uncertainties, and receives program decision-authority disposition.
Key dependencies	1.3.2.1 through 1.3.2.4, architecture trades, transportation constraints, and branch-owner concurrence.
Accountable function	Site Selection Integration Lead
Supporting functions	Program Director, Systems Engineering, Transportation, Habitat, Data Center, Resilience
Primary risk	A premature or poorly integrated site choice creates cross-program cost, safety, and performance impacts.
Decision gate	Lunar Site Characterization Review
Authorization status	Candidate work package for planning. Budget, schedule, named owner, and execution authority are not yet baselined.

Control note: The accountable function becomes responsible for scope, cost, schedule, risk, interfaces, and completion evidence only after formal baseline authorization.

1.3.3.1 Surface Communications and Navigation Architecture

Parent element 1.3.3 Surface Communications and Navigation Setup

Level 4 Work Package Definition	
Work package outcome	Approved surface-network architecture and interface baseline for communications, positioning, navigation, and timing services.
Included work	Coverage requirements, link budgets, topology, frequencies, timing, interfaces, cybersecurity, availability, latency, degraded modes, and growth provisions.
Excluded work	Earth relay network ownership and detailed end-user system design.
Acceptance evidence	Architecture analyses demonstrate required coverage, performance, interoperability, and fault tolerance for approved mission scenarios.
Key dependencies	1.3.1 mission requirements, 1.3.2 terrain data, Earth and lunar relay assumptions, and user-service demands.
Accountable function	Surface Network Architecture Lead
Supporting functions	Communications, Navigation, Cybersecurity, Robotics, Mission Operations
Primary risk	Terrain masking, interference, or inadequate redundancy leaves critical operations without service.
Decision gate	Surface Network Readiness Review
Authorization status	Candidate work package for planning. Budget, schedule, named owner, and execution authority are not yet baselined.

Control note: The accountable function becomes responsible for scope, cost, schedule, risk, interfaces, and completion evidence only after formal baseline authorization.

1.3.3.2 Relay and Base Station Deployment

Parent element 1.3.3 Surface Communications and Navigation Setup

Level 4 Work Package Definition	
Work package outcome	Installed and powered relay nodes, base stations, antennas, and supporting field equipment at controlled locations.
Included work	Deployment locations, transport, placement, anchoring, power connection, antenna alignment, initial configuration, physical protection, and installation evidence.
Excluded work	Design and operation of external orbital relays or shared power generation assets.
Acceptance evidence	All stations are installed at surveyed locations, physically stable, powered, configured, and free of critical installation discrepancies.
Key dependencies	1.3.3.1, 1.3.2.4 hazard maps, 1.3.4 prepared locations, common power availability, and equipment delivery.
Accountable function	Surface Network Deployment Lead
Supporting functions	Robotics, Power, Site Preparation, Logistics, Quality
Primary risk	Installation error, terrain obstruction, or power unavailability prevents planned coverage.
Decision gate	Surface Network Readiness Review
Authorization status	Candidate work package for planning. Budget, schedule, named owner, and execution authority are not yet baselined.

Control note: The accountable function becomes responsible for scope, cost, schedule, risk, interfaces, and completion evidence only after formal baseline authorization.

1.3.3.3 Position Navigation and Timing Establishment

Parent element 1.3.3 Surface Communications and Navigation Setup

Level 4 Work Package Definition	
Work package outcome	Operational local position, navigation, and timing reference supporting autonomous surface activities.
Included work	Reference points, beacons, time distribution, calibration, coordinate frame alignment, localization performance, integrity monitoring, and recovery procedures.
Excluded work	Space-based navigation constellation design and individual vehicle navigation implementation.
Acceptance evidence	Position, navigation, and timing accuracy, availability, integrity, and recovery performance meet approved thresholds across required work zones.
Key dependencies	1.3.3.1, 1.3.3.2, site survey control points, robotic navigation interfaces, and timing source.
Accountable function	Surface Navigation Lead
Supporting functions	Survey, GN&C, Communications, Software, Mission Operations
Primary risk	Position drift or inconsistent reference frames lead to navigation errors and installation misalignment.
Decision gate	Surface Network Readiness Review
Authorization status	Candidate work package for planning. Budget, schedule, named owner, and execution authority are not yet baselined.

Control note: The accountable function becomes responsible for scope, cost, schedule, risk, interfaces, and completion evidence only after formal baseline authorization.

1.3.3.4 Network Integration Cybersecurity and Contingency

Parent element 1.3.3 Surface Communications and Navigation Setup

Level 4 Work Package Definition	
Work package outcome	Integrated network services with controlled access, command authority, data routing, monitoring, and contingency response.
Included work	Identity and access, encryption, command authorization, network management, logging, intrusion detection, alternate paths, loss-of-link response, and restoration procedures.
Excluded work	Enterprise cybersecurity policy ownership and mission-system application security outside the surface network.
Acceptance evidence	Authorized users and systems exchange command and telemetry securely, contingency paths function, and identified cyber findings are dispositioned.
Key dependencies	1.3.3.1 through 1.3.3.3, cybersecurity requirements, operations concept, and external-network interfaces.
Accountable function	Surface Network Operations Lead
Supporting functions	Cybersecurity, Mission Operations, Autonomy, Ground Systems, Safety
Primary risk	Unauthorized command, data loss, or common network failure interrupts autonomous operations.
Decision gate	Surface Network Readiness Review
Authorization status	Candidate work package for planning. Budget, schedule, named owner, and execution authority are not yet baselined.

Control note: The accountable function becomes responsible for scope, cost, schedule, risk, interfaces, and completion evidence only after formal baseline authorization.

1.3.3.5 Coverage Performance Verification and Handover

Parent element 1.3.3 Surface Communications and Navigation Setup

Level 4 Work Package Definition	
Work package outcome	Verified network coverage and performance baseline with an accepted operations handover package.
Included work	Coverage surveys, throughput, latency, availability, navigation accuracy, failover demonstrations, discrepancy closure, as-built configuration, procedures, and training evidence.
Excluded work	Long-term service optimization after operational acceptance.
Acceptance evidence	End-to-end performance satisfies approved requirements in nominal and degraded scenarios, and the receiving operations function accepts the controlled baseline.
Key dependencies	1.3.3.2 through 1.3.3.4, operational users, test instrumentation, and configuration records.
Accountable function	Surface Network Verification Lead
Supporting functions	Test, Quality, Mission Operations, Robotics, Configuration Management
Primary risk	Undetected coverage gaps or incomplete handover cause failures during site preparation and installation.
Decision gate	Surface Network Readiness Review
Authorization status	Candidate work package for planning. Budget, schedule, named owner, and execution authority are not yet baselined.

Control note: The accountable function becomes responsible for scope, cost, schedule, risk, interfaces, and completion evidence only after formal baseline authorization.

1.3.4.1 Site Preparation Plan and Equipment Readiness

Parent element 1.3.4 Site Preparation and Grading

Level 4 Work Package Definition	
Work package outcome	Approved site-preparation execution plan and verified readiness of robotic equipment, tools, and consumables.
Included work	Work sequence, production rates, resource estimates, equipment assignments, tool condition, maintenance, energy demand, safe states, hold points, and contingency methods.
Excluded work	Infrastructure design and general robotic fleet qualification.
Acceptance evidence	The plan is resource loaded at the planning level, interfaces are confirmed, required equipment is ready, and entry criteria are approved.
Key dependencies	1.3.2 site baseline, 1.3.1 fleet readiness, 1.3.3 network readiness, infrastructure layout, and common power plan.
Accountable function	Site Preparation Manager
Supporting functions	Robotics, Construction, Program Controls, Safety, Logistics, Power
Primary risk	Unrealistic production assumptions or unavailable equipment delay the entire surface sequence.
Decision gate	Prepared Site Acceptance Review
Authorization status	Candidate work package for planning. Budget, schedule, named owner, and execution authority are not yet baselined.

Control note: The accountable function becomes responsible for scope, cost, schedule, risk, interfaces, and completion evidence only after formal baseline authorization.

1.3.4.2 Work Zone Demarcation and Hazard Controls

Parent element 1.3.4 Site Preparation and Grading

Level 4 Work Package Definition	
Work package outcome	Established work zones, exclusion areas, routes, control points, and operational hazard controls.
Included work	Boundary marking, coordinate control, traffic rules, keep-out zones, simultaneous-operations controls, emergency routes, command constraints, and verification.
Excluded work	Permanent facility safety systems and crew operational procedures.
Acceptance evidence	All controlled zones and routes match the approved site plan, are visible to autonomous systems, and are accepted before earthmoving begins.
Key dependencies	1.3.2.4 hazard maps, 1.3.3 navigation service, 1.3.4.1 execution plan, and safety analysis.
Accountable function	Surface Safety and Work Control Lead
Supporting functions	Survey, Robotics, Mission Operations, Construction, Safety
Primary risk	Zone ambiguity or conflicting operations cause collision, damage, or unsafe access.
Decision gate	Prepared Site Acceptance Review
Authorization status	Candidate work package for planning. Budget, schedule, named owner, and execution authority are not yet baselined.

Control note: The accountable function becomes responsible for scope, cost, schedule, risk, interfaces, and completion evidence only after formal baseline authorization.

1.3.4.3 Obstacle Removal and Surface Clearing

Parent element 1.3.4 Site Preparation and Grading

Level 4 Work Package Definition	
Work package outcome	Cleared work areas and travel corridors with controlled disposition of boulders, loose material, and identified hazards.
Included work	Obstacle removal, relocation zones, debris handling, surface clearing, dust controls, progress measurement, exception handling, and as-performed records.
Excluded work	Bulk grading, compaction, and final foundation preparation.
Acceptance evidence	Specified areas meet obstacle-size, clearance, access, and debris-disposition criteria with no unresolved critical hazards.
Key dependencies	1.3.4.1, 1.3.4.2, robotic equipment readiness, disposal locations, and environmental constraints.
Accountable function	Robotic Construction Lead
Supporting functions	Site Preparation, Safety, Survey, Maintenance, Environmental Monitoring
Primary risk	Unexpected obstacles, equipment wear, or dust reduce production and damage critical assets.
Decision gate	Prepared Site Acceptance Review
Authorization status	Candidate work package for planning. Budget, schedule, named owner, and execution authority are not yet baselined.

Control note: The accountable function becomes responsible for scope, cost, schedule, risk, interfaces, and completion evidence only after formal baseline authorization.

1.3.4.4 Grading Compaction and Foundation Preparation

Parent element 1.3.4 Site Preparation and Grading

Level 4 Work Package Definition	
Work package outcome	Graded, compacted, and dimensionally controlled pads, berms, trenches, and foundations ready for installation.
Included work	Cut and fill, leveling, compaction, berm and trench formation, regolith conditioning, drainage assumptions, dimensional control, quality sampling, and rework.
Excluded work	Equipment anchoring, structural assembly, and system commissioning.
Acceptance evidence	Prepared features meet approved elevation, slope, flatness, compaction, geometry, and load-bearing criteria with verified survey evidence.
Key dependencies	1.3.4.3, approved infrastructure layouts, geotechnical criteria, survey control, and construction equipment availability.
Accountable function	Lunar Civil Works Lead
Supporting functions	Robotics, Geotechnical, Survey, Quality, Infrastructure Owners
Primary risk	Insufficient compaction or dimensional error drives settlement, misalignment, or installation rework.
Decision gate	Prepared Site Acceptance Review
Authorization status	Candidate work package for planning. Budget, schedule, named owner, and execution authority are not yet baselined.

Control note: The accountable function becomes responsible for scope, cost, schedule, risk, interfaces, and completion evidence only after formal baseline authorization.

1.3.4.5 Prepared Site Acceptance Survey and Handover

Parent element 1.3.4 Site Preparation and Grading

Level 4 Work Package Definition	
Work package outcome	Controlled as-built survey, quality record, discrepancy disposition, and handover of prepared areas to installation teams.
Included work	Final survey, acceptance sampling, load-bearing confirmation, configuration records, open-item assessment, nonconformance closure, release status, and handover package.
Excluded work	Installation and operation of the infrastructure placed on accepted areas.
Acceptance evidence	Each area has objective acceptance evidence, approved residual deviations, an as-built baseline, and documented acceptance by the receiving owner.
Key dependencies	1.3.4.2 through 1.3.4.4, acceptance criteria from infrastructure owners, and configuration-management controls.
Accountable function	Prepared Site Acceptance Lead
Supporting functions	Quality, Survey, Systems Engineering, Habitat, Data Center, Shared Infrastructure
Primary risk	Incomplete evidence or unresolved deviations transfer latent defects into installation and commissioning.
Decision gate	Prepared Site Acceptance Review
Authorization status	Candidate work package for planning. Budget, schedule, named owner, and execution authority are not yet baselined.

Control note: The accountable function becomes responsible for scope, cost, schedule, risk, interfaces, and completion evidence only after formal baseline authorization.

1.3.5.1 Deployment Sequence and Robotic Assembly Planning

Parent element 1.3.5 Infrastructure Deployment and Assembly

Level 4 Work Package Definition	
Work package outcome	Integrated deployment and assembly sequence aligned to cargo arrival, prepared-site availability, utility interfaces, and commissioning priorities.
Included work	Installation sequence, lift and handling concepts, robotic task plans, hold points, simultaneous operations, temporary support, resources, contingency paths, and schedule interfaces.
Excluded work	Design authority for installed systems and detailed transportation mission planning.
Acceptance evidence	The sequence is logically linked, resource feasible, interface approved, and demonstrates a recoverable path for critical installation scenarios.
Key dependencies	1.3.4 accepted sites, cargo manifest, system installation requirements, robotic capabilities, and common-service availability.
Accountable function	Surface Installation Manager
Supporting functions	Program Controls, Robotics, Logistics, Habitat, Data Center, Shared Infrastructure
Primary risk	Sequence conflicts, inaccessible interfaces, or missing enabling services cause installation deadlock.
Decision gate	Infrastructure Installation Readiness Review
Authorization status	Candidate work package for planning. Budget, schedule, named owner, and execution authority are not yet baselined.

Control note: The accountable function becomes responsible for scope, cost, schedule, risk, interfaces, and completion evidence only after formal baseline authorization.

1.3.5.2 Cargo Offload Transport and Staging

Parent element 1.3.5 Infrastructure Deployment and Assembly

Level 4 Work Package Definition	
Work package outcome	Safely offloaded, identified, inspected, transported, and staged infrastructure cargo at controlled locations.
Included work	Offload interfaces, lifting and towing, receiving inspection, inventory control, staging layout, temporary protection, damage reporting, and delivery records.
Excluded work	Descent and landing operations before surface release and final equipment installation.
Acceptance evidence	Cargo is accounted for, free of unresolved critical damage, staged in the required sequence, and released to installation under configuration control.
Key dependencies	Element 1.2 surface delivery, 1.3.2.4 routes, 1.3.4 accepted staging areas, handling equipment, and cargo documentation.
Accountable function	Lunar Logistics Lead
Supporting functions	Robotics, Quality, Transportation, Configuration Management, Infrastructure Owners
Primary risk	Landing damage, misidentification, or inaccessible staging disrupts the installation sequence.
Decision gate	Infrastructure Installation Readiness Review
Authorization status	Candidate work package for planning. Budget, schedule, named owner, and execution authority are not yet baselined.

Control note: The accountable function becomes responsible for scope, cost, schedule, risk, interfaces, and completion evidence only after formal baseline authorization.

1.3.5.3 Utility Pathway and Interface Installation

Parent element 1.3.5 Infrastructure Deployment and Assembly

Level 4 Work Package Definition	
Work package outcome	Installed pathways, conduits, connectors, distribution interfaces, and service tie points required for later system connection.
Included work	Routing, trench and conduit placement, connector installation, cable or line protection, labeling, isolation, continuity checks, interface inspection, and as-built records.
Excluded work	Generation and operation of common utilities and internal system distribution beyond the interface boundary.
Acceptance evidence	All utility pathways and tie points meet routing, protection, identification, continuity, isolation, and interface-control requirements.
Key dependencies	1.3.4.4 prepared features, approved interface-control documents, delivered hardware, and installation sequence.
Accountable function	Surface Utilities Installation Lead
Supporting functions	Power, Communications, Thermal, Robotics, Quality, Configuration Management
Primary risk	Incorrect routing or interface damage creates common-service failures or costly rework.
Decision gate	Infrastructure Installation Readiness Review
Authorization status	Candidate work package for planning. Budget, schedule, named owner, and execution authority are not yet baselined.

Control note: The accountable function becomes responsible for scope, cost, schedule, risk, interfaces, and completion evidence only after formal baseline authorization.

1.3.5.4 Equipment Placement Anchoring and Assembly

Parent element 1.3.5 Infrastructure Deployment and Assembly

Level 4 Work Package Definition	
Work package outcome	Installed equipment positioned, aligned, anchored, mechanically assembled, and protected in accordance with controlled requirements.
Included work	Positioning, leveling, anchoring, module joining, mechanical connections, deployment mechanisms, torque or preload verification, protective features, and inspection evidence.
Excluded work	Internal activation, software commissioning, and operational acceptance of the installed systems.
Acceptance evidence	Equipment location, orientation, alignment, anchoring, assembly, and physical protection meet approved tolerances with no unresolved critical discrepancies.
Key dependencies	1.3.5.1 through 1.3.5.3, accepted foundations, handling fixtures, installation procedures, and design-authority support.
Accountable function	Robotic Assembly Lead
Supporting functions	Structures, Mechanical, Robotics, Quality, Habitat, Data Center
Primary risk	Alignment error, anchoring failure, or robotic handling damage prevents commissioning.
Decision gate	Infrastructure Installation Readiness Review
Authorization status	Candidate work package for planning. Budget, schedule, named owner, and execution authority are not yet baselined.

Control note: The accountable function becomes responsible for scope, cost, schedule, risk, interfaces, and completion evidence only after formal baseline authorization.

1.3.5.5 Installation Verification and System Handover

Parent element 1.3.5 Infrastructure Deployment and Assembly

Level 4 Work Package Definition	
Work package outcome	Verified installation baseline and formal handover package for system commissioning by the owning branches.
Included work	As-built configuration, interface verification, continuity and leak checks where applicable, visual inspection, discrepancy closure, preservation status, turnover records, and owner acceptance.
Excluded work	Functional commissioning and operational performance validation owned by elements 1.4 through 1.6.
Acceptance evidence	The owning branch accepts complete installation evidence, controlled as-built records, verified interfaces, and an agreed list of noncritical remaining work.
Key dependencies	1.3.5.2 through 1.3.5.4, system acceptance criteria, configuration records, quality evidence, and commissioning prerequisites.
Accountable function	Installation Verification Lead
Supporting functions	Quality, Configuration Management, Systems Engineering, Receiving Branch Owners
Primary risk	Incomplete turnover data or hidden installation defects delay commissioning and blur accountability.
Decision gate	Infrastructure Installation Readiness Review
Authorization status	Candidate work package for planning. Budget, schedule, named owner, and execution authority are not yet baselined.

Control note: The accountable function becomes responsible for scope, cost, schedule, risk, interfaces, and completion evidence only after formal baseline authorization.

1.3.6.1 Inspection and Condition Baseline

Parent element 1.3.6 Inspection Maintenance and Remote Readiness Verification

Level 4 Work Package Definition	
Work package outcome	Controlled initial condition baseline for robotic assets, prepared sites, installed infrastructure, and critical interfaces.
Included work	Inspection plan, imagery, sensor readings, dimensional checks, wear indicators, contamination, damage classification, condition limits, and baseline records.
Excluded work	Routine long-term inspections after formal operational handover.
Acceptance evidence	Baseline evidence covers all critical assets and interfaces, identifies deviations, and supports repeatable future condition comparisons.
Key dependencies	1.3.4 and 1.3.5 acceptance records, inspection access, sensor calibration, and configuration baseline.
Accountable function	Remote Inspection Lead
Supporting functions	Quality, Robotics, Structures, Utilities, Data Management
Primary risk	Incomplete baselines prevent early degradation detection and defensible readiness decisions.
Decision gate	Uncrewed Surface Readiness Review
Authorization status	Candidate work package for planning. Budget, schedule, named owner, and execution authority are not yet baselined.

Control note: The accountable function becomes responsible for scope, cost, schedule, risk, interfaces, and completion evidence only after formal baseline authorization.

1.3.6.2 Preventive Maintenance Spares and Repair Readiness

Parent element 1.3.6 Inspection Maintenance and Remote Readiness Verification

Level 4 Work Package Definition	
Work package outcome	Approved preventive-maintenance, repair, tooling, consumables, and spares capability for the uncrewed operating period.
Included work	Maintenance intervals, condition triggers, spare prioritization, storage, robotic repair methods, tool access, consumables, cannibalization rules, and replenishment assumptions.
Excluded work	Crew-performed maintenance after human arrival and supplier production planning.
Acceptance evidence	Critical maintenance tasks have feasible remote methods, required resources, verified access, and controlled spare or contingency coverage.
Key dependencies	1.3.6.1 condition baseline, reliability data, logistics constraints, robotic tooling, and asset-owner maintenance requirements.
Accountable function	Surface Maintenance Lead
Supporting functions	Reliability, Logistics, Robotics, Infrastructure Owners, Program Controls
Primary risk	A single unavailable spare, tool, or repair method causes extended loss of a critical capability.
Decision gate	Uncrewed Surface Readiness Review
Authorization status	Candidate work package for planning. Budget, schedule, named owner, and execution authority are not yet baselined.

Control note: The accountable function becomes responsible for scope, cost, schedule, risk, interfaces, and completion evidence only after formal baseline authorization.

1.3.6.3 Remote Command Diagnostics and Recovery Validation

Parent element 1.3.6 Inspection Maintenance and Remote Readiness Verification

Level 4 Work Package Definition	
Work package outcome	Verified remote monitoring, command, diagnostic, isolation, safe-state, and recovery capability across the surface system.
Included work	Telemetry, health limits, command paths, diagnostic logic, fault isolation, remote reset, alternate modes, communication-loss behavior, and recovery demonstrations.
Excluded work	Detailed internal fault management owned by each system design authority.
Acceptance evidence	Representative faults are detected, isolated, communicated, and recovered or placed in an approved safe state within defined thresholds.
Key dependencies	1.3.3 network acceptance, asset telemetry and command interfaces, operations procedures, and autonomy functions.
Accountable function	Remote Operations Readiness Lead
Supporting functions	Mission Operations, Communications, Cybersecurity, Autonomy, System Owners
Primary risk	Insufficient observability or remote authority allows recoverable faults to become extended outages.
Decision gate	Uncrewed Surface Readiness Review
Authorization status	Candidate work package for planning. Budget, schedule, named owner, and execution authority are not yet baselined.

Control note: The accountable function becomes responsible for scope, cost, schedule, risk, interfaces, and completion evidence only after formal baseline authorization.

1.3.6.4 Anomaly Resolution and Configuration Closure

Parent element 1.3.6 Inspection Maintenance and Remote Readiness Verification

Level 4 Work Package Definition	
Work package outcome	Controlled closure or formal risk acceptance of anomalies, discrepancies, temporary configurations, and open work before readiness review.
Included work	Anomaly log, root-cause assessment, corrective action, retest, waiver and deviation control, configuration reconciliation, trend review, and closure evidence.
Excluded work	New design development not approved through change control and routine operational issue management after handover.
Acceptance evidence	All safety-critical and mission-critical anomalies are closed; remaining items have approved ownership, risk disposition, and completion plans.
Key dependencies	1.3.6.1 through 1.3.6.3, quality records, change control, risk register, and branch-owner concurrence.
Accountable function	Mission Assurance and Configuration Lead
Supporting functions	Quality, Systems Engineering, Program Controls, Safety, Infrastructure Owners
Primary risk	Uncontrolled open work or configuration ambiguity invalidates readiness evidence.
Decision gate	Uncrewed Surface Readiness Review
Authorization status	Candidate work package for planning. Budget, schedule, named owner, and execution authority are not yet baselined.

Control note: The accountable function becomes responsible for scope, cost, schedule, risk, interfaces, and completion evidence only after formal baseline authorization.

1.3.6.5 Uncrewed Readiness Demonstration and Crew Arrival Recommendation

Parent element 1.3.6 Inspection Maintenance and Remote Readiness Verification

Level 4 Work Package Definition	
Work package outcome	Integrated uncrewed operations demonstration and evidence-based recommendation on readiness for the next program phase and crew arrival.
Included work	End-to-end scenarios, sustained operations, degraded modes, emergency response, maintenance demonstrations, performance trending, residual-risk assessment, decision package, and lessons learned.
Excluded work	Authorization of crew launch or human-rating certification controlled outside this work package.
Acceptance evidence	The integrated demonstration meets approved success criteria, residual risks are documented, and the decision authority accepts or conditions the readiness recommendation.
Key dependencies	1.3.6.1 through 1.3.6.4, accepted habitat and common-service status, mission scenarios, and program risk thresholds.
Accountable function	Integrated Readiness Lead
Supporting functions	Program Director, Mission Operations, Safety, Habitat, Data Center, Resilience
Primary risk	A narrow demonstration overlooks cross-system or time-dependent failures that threaten crew safety or mission continuity.
Decision gate	Uncrewed Surface Readiness Review
Authorization status	Candidate work package for planning. Budget, schedule, named owner, and execution authority are not yet baselined.

Control note: The accountable function becomes responsible for scope, cost, schedule, risk, interfaces, and completion evidence only after formal baseline authorization.

Review and Baseline Actions

Required management decisions before detailed execution planning

Recommended review sequence

- Assign each Level 3 branch to a responsible functional lead for technical validation.
- Confirm that every Level 4 package produces a unique deliverable and has no hidden overlap with elements 1.2 and 1.4 through 1.6.
- Select the packages that require formal work-package control and identify any activities that should remain planning packages or level-of-effort support.
- Convert acceptance statements into measurable thresholds linked to requirements and verification evidence.
- Establish the integrated logic network across survey, communications, site preparation, installation, inspection, and readiness gates.
- Assign named owners, resources, budgets, dates, and risk responses only after scope and interface approval.

Recommended next artifact

Develop the Responsibility Assignment Matrix and preliminary integrated milestone network for WBS element 1.3. These artifacts should identify accountable and supporting organizations, decision authorities, cross-branch interfaces, major predecessors, and the path to the Uncrewed Surface Readiness Review.

Revision record: Version 0.1, 22 September 2026. Initial Level 4 decomposition of WBS element 1.3 into 30 candidate work packages.